



Name: \_\_\_\_\_

Teacher: \_\_\_\_\_

Class: \_\_\_\_\_

**FORT STREET HIGH SCHOOL**

**2009**

**PRELIMINARY SCHOOL CERTIFICATE COURSE  
ASSESSMENT TASK 1**

# Mathematics

**TIME ALLOWED: 1 HOUR**

| Outcomes Assessed                                                                                                        | Questions | Marks |
|--------------------------------------------------------------------------------------------------------------------------|-----------|-------|
| Performs routine arithmetic manipulation involving surds and simple rational expressions                                 | 1         |       |
| Chooses and applies appropriate algebraic techniques to simplify algebraic expressions                                   | 2         |       |
| Solves problems involving equations                                                                                      | 3         |       |
| Chooses and applies appropriate geometric techniques and provides appropriate reasoning to support conclusions in proofs | 4         |       |

| Question | 1   | 2   | 3   | 4   | Total | % |
|----------|-----|-----|-----|-----|-------|---|
| Marks    | /16 | /16 | /16 | /16 | /64   |   |

## Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used
- Each new question is to be started on a new booklet

**Question One:** (16 marks) **Start a new booklet**

- a) Find  $|-2.5| - |8|$  [1]
- b) Simplify  $1\frac{3}{5} \div 2\frac{2}{3}$  [1]
- c) Find  $\frac{1}{2.82 - 1.97}$  (correct to 3 significant figures) [2]
- d) Calculate  $6.31^3 \times \sqrt{4.301} \div 9.38$  (correct to 2 decimal places) [2]
- e) Simplify  $\sqrt{24} + 3\sqrt{48} - 2\sqrt{54} + 4\sqrt{75}$  [2]
- f) Calculate  $\sqrt{\frac{305^2 + 4071}{6 \div 8.6^3}}$  (give you answer in scientific notation to 4 significant figures) [2]
- g) Expand  $(2\sqrt{2} - 1)(\sqrt{2} + 3)$  [2]
- h) Evaluate  $\frac{a^3 b^4}{c^2}$  if  $a = \left(\frac{2}{3}\right)^2$   $b = \left(\frac{3}{4}\right)^3$   $c = \left(\frac{3}{8}\right)^4$ , leaving your answer as an improper fraction [2]
- i) Rationalize the denominator:  $\frac{2\sqrt{3} + 3}{1 - \sqrt{3}}$  [2]

**Question Two:** (16 marks) **Start a new booklet**

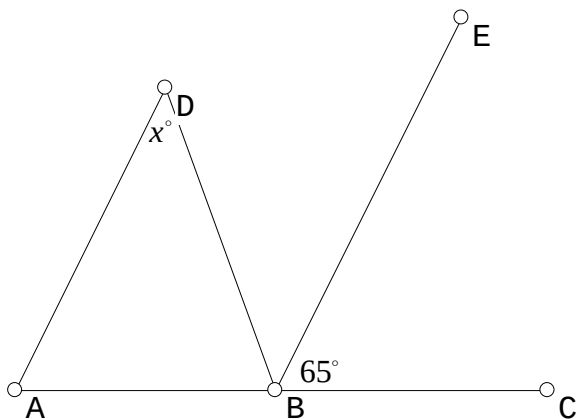
- a) Expand  $(x - 2)(2x + 5)$  [1]
- b) Factorize  $x^2 + 14x - 32$  [1]
- c) Simplify  $6x^2 + 5x - 2x^2 + 4x - 3$  [2]
- d) Simplify  $\frac{a+2}{3} - \frac{a}{4}$  [2]
- e) Fully factorize  $x^4 - 4x^2$  [2]
- f) Simplify  $(2x+1)^2 - (x-2)^2$  [2]
- g) Given  $S = \frac{a(1-r^n)}{1-r}$ , find  $S$  when  $a = 2$ ,  $r = \frac{1}{3}$ ,  $n = 12$  [2]
- h) Simplify  $\frac{p^2 - 5p + 4}{p^2 - 1}$  [2]
- i) Factorise  $27x^3 + 125$  [2]

**Question Three:** (16 marks) **Start a new booklet**

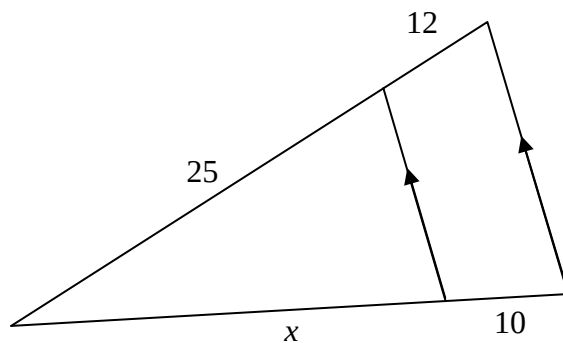
- a) Solve  $-2(3x+1)=8$  [1]
- b) Draw a neat number-line to show the solution to  $\frac{x+5}{3}+3>4$  [2]
- c) Solve  $x^2-3x-10=0$  [2]
- d) Use the quadratic formula to find the roots of  $x^2-6x+7=0$  in surd form. [2]
- e) Solve  $|x-3|=2x-1$  [3]
- f) Solve  $y=x^2-4x-1$  and  $y=2x+3$  simultaneously. [4]
- g) A regular polygon has an interior angle of  $156^\circ$ . Find how many sides it has. [2]

**Question four:** (16 marks) **Start a new booklet**

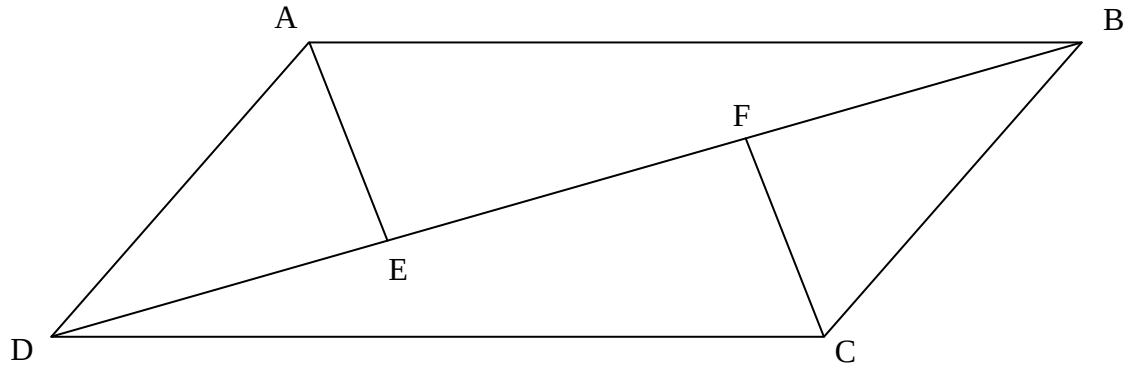
- a) Find the value of  $x$  in the diagrams below
- i.  $AD = DB$  and  $AD \parallel BE$ . Give reasons. [3]



- ii. [2]

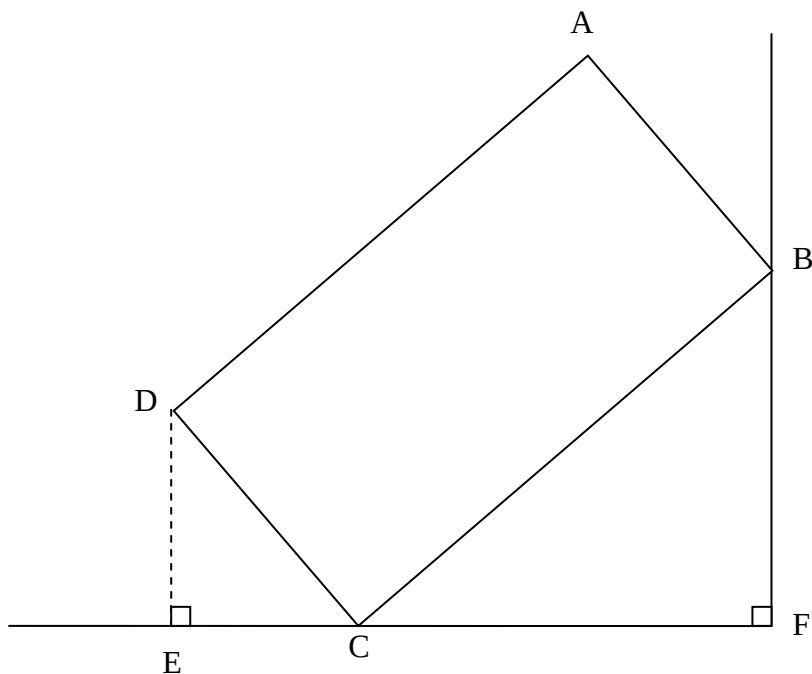


- b)  $ABCD$  is a parallelogram, with  $AE$  and  $CF$  drawn perpendicular to the diagonal  $BD$ .



- i. Prove  $\triangle AEB \equiv \triangle DFC$  [4]
- ii. Hence prove  $AE = FC$  [2]

- c)  $ABCD$  is a rectangular block 10cm by 6cm leaning against a vertical wall. Point  $D$  is 5cm above the ground.



- i. Prove  $\triangle CED$  and  $\triangle FCB$  are similar. [3]
- ii. Find, in simplest surd form, the height of point  $B$  above the ground. [2]

**Question One:** (16 marks) **Start a new booklet**

a)  $|-2.5| - |8|$   
 $= 2.5 - 8$   
 $= -5.5$

[1]

Marking

Comments

1 answer

b)  $1\frac{3}{5} \div 2\frac{2}{3}$   
 $= \frac{8}{5} \times \frac{3}{8}$   
 $= \frac{3}{5}$

[1]

1 answer

c)  $\frac{1}{2.82 - 1.97}$   
 $= \frac{1}{0.85}$   
 $= 1.176470588$   
 $= 1.18$

[2]

1 answer

1 3 sig figs correct

§

d)  $6.31^3 \times \sqrt{4.301} \div 9.38$   
 $= 55.54819617$   
 $= 55.55$

[2]

1 answer

1 2 dp correct

§

e)  $\sqrt{24} + 3\sqrt{48} - 2\sqrt{54} + 4\sqrt{75}$   
 $= \sqrt{4 \times 6} + 3\sqrt{16 \times 3} - 2\sqrt{9 \times 6} + 4\sqrt{25 \times 3}$   
 $= 2\sqrt{6} + 3 \times 4\sqrt{3} - 2 \times 3\sqrt{6} + 4 \times 5\sqrt{3}$   
 $= 32\sqrt{3} - 4\sqrt{6}$

[2]

1 surds reduced correctly

1 combined like surds correctly

- students had trouble reducing  $\sqrt{48}$  and  $\sqrt{75}$  to multiples of  $\sqrt{3}$

f)  $\sqrt{\frac{305^2 + 4071}{6 \div 8.6^3}}$   
 $= 3208.283377$   
 $= 3.208 \times 10^3$

[2]

1 answer

1 sci. not. correct

§

g)  $(2\sqrt{2} - 1)(\sqrt{2} + 3)$   
 $= \sqrt{2}(2\sqrt{2} - 1) + 3(2\sqrt{2} - 1)$   
 $= 4 - \sqrt{2} + 6\sqrt{2} - 3$   
 $= 1 + 5\sqrt{2}$

[2]

1 correct expansion

1 collected like terms correctly

- as with numerator in i) below, many had problems expanding with surds.

h)

$$a = \left(\frac{2}{3}\right)^2 \quad b = \left(\frac{3}{4}\right)^3 \quad c = \left(\frac{3}{8}\right)^4$$

$$\therefore \frac{a^3 b^4}{c^2}$$

$$= \frac{\left(\left(\frac{2}{3}\right)^2\right)^3 \times \left(\left(\frac{3}{4}\right)^3\right)^4}{\left(\left(\frac{3}{8}\right)^4\right)^2}$$

$$= \frac{\left(\frac{2}{3}\right)^6 \times \left(\frac{3}{4}\right)^{12}}{\left(\frac{3}{8}\right)^8}$$

$$= \frac{2^6}{3^6} \times \frac{3^{12}}{2^{24}} \times \frac{2^{24}}{3^8}$$

$$= \frac{2^6}{3^2}$$

$$= \frac{64}{9}$$

i)

$$\frac{2\sqrt{3}+3}{1-\sqrt{3}}$$

$$= \frac{2\sqrt{3}+3}{1-\sqrt{3}} \times \frac{1+\sqrt{3}}{1+\sqrt{3}}$$

$$= \frac{(2\sqrt{3}+3) + \sqrt{3}(2\sqrt{3}+3)}{1^2 - (\sqrt{3})^2}$$

$$= \frac{2\sqrt{3}+3+6+3\sqrt{3}}{1-3}$$

$$= -\frac{(9+5\sqrt{3})}{2}$$

[2]

Marking

Comments

1 powers resolved correctly

1 answer

- many had problems leaving or converting to powers of 2 and 3, so had problems resolving as a fraction

[2]

1 correct conjugate

1 answer

- those losing marks on this question generally did not have the conjugate correct.

**Question Two:** (16 marks) **Start a new booklet**

a)  $(x-2)(2x+5)$  [1]

$$= x(2x+5) - 2(2x+5)$$

$$= 2x^2 + 5x - 4x - 10$$

$$= 2x^2 + x - 10$$

b)  $x^2 + 14x - 32$  [1]

$$= (x+16)(x-2)$$

c)  $6x^2 + 5x - 2x^2 + 4x - 3$  [2]

$$= 4x^2 + 9x - 3$$

d) Simplify  $\frac{a+2}{3} - \frac{a}{4}$  [2]

$$= \frac{4(a+2) - 3a}{12}$$

$$= \frac{4a + 8 - 3a}{12}$$

$$= \frac{a+8}{12}$$

e)  $x^4 - 4x^2$  [2]

$$= x^2(x^2 - 4)$$

$$= x^2(x+2)(x-2)$$

f)  $(2x+1)^2 - (x-2)^2$  [2]

$$= 4x^2 + 4x + 1 - (x^2 - 4x + 4)$$

$$= 3x^2 + 8x - 3$$

g)  $a = 2, \quad r = \frac{1}{3}, \quad n = 12$  [2]

$$S = \frac{a(1-r^n)}{1-r}$$

$$= \frac{2\left(1 - \left(\frac{1}{3}\right)^{12}\right)}{1 - \frac{1}{3}}$$

$$= 2\left(1 - \frac{1}{3^{12}}\right) \times \frac{3}{2}$$

$$= 3\left(\frac{3^{12} - 1}{3^{12}}\right)$$

$$= 2.999994355$$

## Marking

## Comments

1 answer

1 answer

1  $x^2$  term correct1  $x$  term correct

1 correct cross multiplication

1 answer

1  $x^2$  factored

1 difference of squares correct

1 expansions

1 answer

1 substitutions &amp; powers correct

1 answer

Note: a), b) and c) are EXPRESSIONS - too many treated them as equations and answered as  $x = \dots$

Note: this is not a factorising question!

- many missed the  $x^2$  factor.

- many made errors subtracting the expansion; brackets often missed

Note: the question does not ask for a number of decimal places; in these cases, quote all from your calculator; an answer of 3 is incorrect (no mark deducted if the answer show was subsequently rounded to 3)

$$\text{h) } \frac{p^2 - 5p + 4}{p^2 - 1}$$

$$= \frac{(p-4)(p-1)}{(p+1)(p-1)}$$

$$= \frac{(p-4)}{(p+1)}$$

$$\text{i) } 27x^3 + 125$$

$$= (3x+5)(9x^2 - 15x + 25)$$

[2]

Marking

Comments

① factors correct

① answer

[2]

① cube root bracket

① squares bracket

- this pattern  
poorly known

**Question Three:** (16 marks) **Start a new booklet**

$$\text{a) } -2(3x+1) = 8$$

$$-6x - 2 = 8$$

$$-10 = 6x$$

$$x = \frac{-5}{3}$$

[1]

- simplest form!  
 $\frac{-10}{6}$  accepted  
(this time)...

① answer with  
working

$$\text{b) } \frac{x+5}{3} + 3 > 4$$

$$\frac{x+5}{3} > 1$$

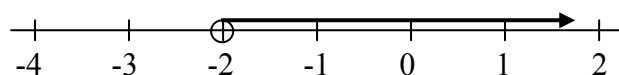
$$x+5 > 3$$

$$x > -2$$

[2]

Note: a) and c)  
are  
EQUATIONS –  
they DO need  
an answer of  
 $x = \dots$

① resolved inequality



① number line correct

$$\text{c) } x^2 - 3x - 10 = 0$$

$$(x-5)(x+2) = 0$$

$$x = 5, -2$$

[2]

① factors correct

① answers

- too many  
stopped at the  
factored form,  
and did not give  
the x values.

d)

$$x^2 - 6x + 7 = 0$$

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{6 \pm \sqrt{(-6)^2 - 4 \times 1 \times 7}}{2 \times 1} \\ &= \frac{6 \pm \sqrt{36 - 28}}{2} \\ &= \frac{6 \pm \sqrt{8}}{2} \\ &= \frac{6 \pm 2\sqrt{2}}{2} \\ &= 3 \pm \sqrt{2} \end{aligned}$$

e)  $|x - 3| = 2x - 1$

$$x - 3 = 2x - 1 \text{ or } -(x - 3) = 2x - 1$$

$$-2 = x \quad 4 = 3x$$

$$x = -2 \quad x = \frac{4}{3}$$

Check:

$$LHS = \left| \frac{4}{3} - 3 \right|$$

$$\begin{aligned} LHS &= |-2 - 3| &= \left| -\frac{5}{3} \right| \\ &= |-5| &= \frac{5}{3} \\ &= 5 &= \frac{5}{3} \end{aligned}$$

$$\begin{aligned} RHS &= 2 \times -2 - 1 \\ &= -5 \end{aligned} \quad \begin{aligned} RHS &= 2 \times \frac{4}{3} - 1 \\ &= \frac{5}{3} \end{aligned}$$

$$\begin{aligned} &\neq LHS \\ &= \frac{5}{3} \\ &= LHS \end{aligned}$$

$\therefore x = \frac{4}{3}$  is the only solution.

[2]

Marking

Comments

1 correct substitution in quadratic formula

- resolving  $\sqrt{8}$  to  $2\sqrt{2}$  an issue for some, leading to errors giving the simplest form of the surd

1 answers

[3]

1 both cases shown

- several students couldn't get two answers for the two cases

1 possible solutions correct

- most common error was not checking the solutions. You MUST do this step every time there is an absolute value = a non-absolute value.

1 checking correct

f)  $y = x^2 - 4x - 1$  and  $y = 2x + 3$ .

$$2x + 3 = x^2 - 4x - 1$$

$$0 = x^2 - 6x - 4$$

$$x = \frac{6 \pm \sqrt{(-6)^2 - 4 \times 1 \times -4}}{2 \times 1}$$

$$= \frac{6 \pm \sqrt{52}}{2}$$

$$= \frac{6 \pm 2\sqrt{13}}{2}$$

$$= 3 \pm \sqrt{13}$$

$$x = 3 + \sqrt{13} \quad x = 3 - \sqrt{13}$$

$$y = 2(3 + \sqrt{13}) + 3 \quad y = 2(3 - \sqrt{13}) + 3$$

$$= 6 + 2\sqrt{13} + 3 \quad = 6 - 2\sqrt{13} + 3$$

$$= 9 + 2\sqrt{13} \quad = 9 - 2\sqrt{13}$$

$\therefore$  Points of intersection are

$$(2 + \sqrt{13}, 9 + 2\sqrt{13}) \quad (2 - \sqrt{13}, 9 - 2\sqrt{13})$$

g) Regular polygon with interior angle of  $156^\circ$ : [2]

$$\theta = \frac{180(n-2)}{n}$$

$$156 = \frac{180(n-2)}{n}$$

$$156n = 180n - 360$$

$$360 = 24n$$

$$n = 15$$

$\therefore$  the polygon has 15 sides.

[4]

## Marking

❶ quadratic correctly formed

❶ quadratic formula substitution correct

❶ X values correct

❶ y values correct

❶ correct substitution in polygon formula

❶ answer

## Comments

- many did not put as quadratic =0, so were unable to proceed

- solving simultaneously means finding the points of intersection – many did not find the y-values.

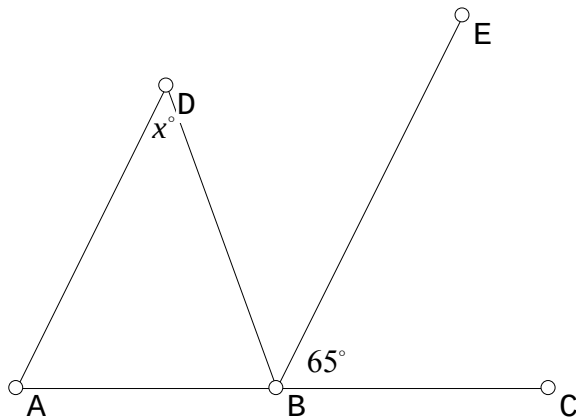
- some students did not use a valid formula and were

**Question Four:** (16 marks) **Start a new booklet**

a) Find the value of  $x$  in the diagrams below

- i.  $AD = DB$  and  $AD \parallel BE$ . Give reasons.

[3]



$$\angle DAB = 65^\circ \quad (\text{cor } \angle \text{'s } AD \parallel BE)$$

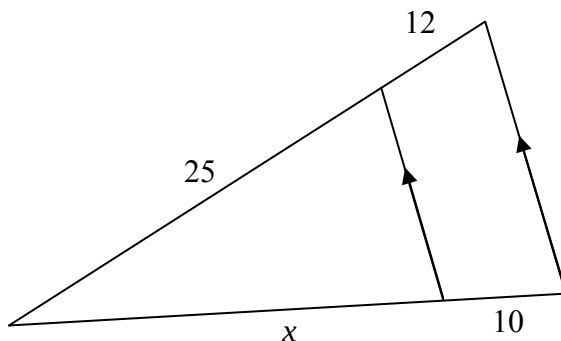
$$\angle ABD = 65^\circ \quad (\text{base } \angle \text{'s } \triangle ABD)$$

$$x + 2 \times 65 = 180 \quad (\triangle \text{Sum})$$

$$x = 50$$

ii.

[2]



From similar triangles:

$$\frac{x}{25} = \frac{10}{12} \quad \text{OR} \quad \frac{x}{25} = \frac{x+10}{25+12}$$

$$12x = 250$$

$$37x = 25x + 250$$

$$x = \frac{125}{6}$$

$$12x = 250$$

$$x = \frac{125}{6}$$

$$x = 20\frac{5}{6}$$

$$x = 20\frac{5}{6}$$

Marking

Comments

1 value with reason

1 value with reason

1 answer with reason

1 correct ratio setup

1 answer

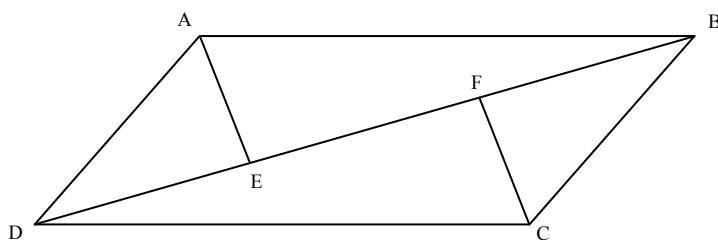
For both parts of a)

- generally not well set out; it was often difficult to follow reasoning.

- references to parallel line properties  
MUST mention the parallel lines in question (no marks deducted – this time!)

- mostly well done; method depended on how students saw the ratios – parallel lines  
OR similar triangles.

- b)  $ABCD$  is a parallelogram, with  $AE$  and  $CF$  drawn perpendicular to the diagonal  $BD$ .



- i. Prove  $\triangle AEB \equiv \triangle DFC$  [4]

In  $\triangle$ 's  $AEB, DFC$

i)  $\angle AEB = \angle CFD (= 90^\circ)$  (given  $AE \perp BD, CF \perp BD$ )

ii)  $\angle EBA = \angle FDC$  (given  $BC \parallel AD, ABCD \parallel \text{gram}$ )

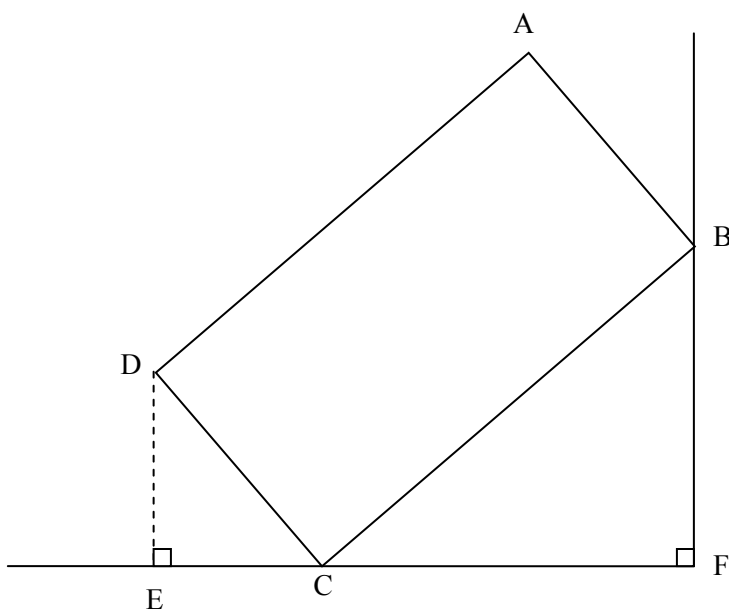
iii)  $AD = BC$  (op. sides of  $\parallel \text{gram } ABCD$ )

$\therefore \triangle AEB \equiv \triangle DFC$  (AAS)

- ii. Hence prove  $AE = FC$  [2]

$\therefore AE = FC$  (cor.  $\angle$ 's cong.  $\triangle$ 's above)

- c)  $ABCD$  is a rectangular block 10cm by 6cm leaning against a vertical wall. Point D is 5cm above the ground.



## Marking

① set up proof

② three congruency components; subtract

① per error

① conclusion with correct reason

① corresponding as reason

① reference to congruent triangles/proof above

## Comments

NOTE:  
GEOMETRY  
PROOFS  
GENERALLY!

These are very formal in structure – while content of a proof may vary, there is a right way in terms of structure (and no other)!

- many omitted the set-up step.

- several students claimed “BD common” and a RHS proof – clearly not using the correct triangles!

i. Prove  $\triangle CED$  and  $\triangle FCB$  are similar. [3]

$$\angle ECD + \angle DCB + \angle BCF = 180^\circ \quad (\text{st. line})$$

but  $\angle DCB = 90^\circ$  ( $ABCD$  a rectangle)

hence  $\angle ECD + \angle BCF = 90^\circ$

so  $\angle ECD = 90^\circ - \angle BCF$

also  $\angle BCF + \angle CBF + \angle BFC = 180^\circ$  ( $\triangle sum$ )

and  $\angle BFC = 90^\circ$  (given  $BF \perp EB$ )

so  $\angle BCF + \angle CBF = 90^\circ$

hence  $\angle CBF = 90^\circ - \angle BCF$

but  $\angle ECD = 90^\circ - \angle BCF$  (shown above)

$$\therefore \angle ECD = \angle CBF$$

$\therefore$  In  $\Delta$ 's DCE, CBF

i)  $\angle ECD = \angle CBF$  (shown above)

$$ii) \angle CED = \angle BFC = 90^\circ \quad (DE \perp EF, BF \perp EF)$$

hence  $\triangle DCE \parallel \triangle CBF$  (equal  $\angle$ 's)

ii. Height of point B. [2]

hence

$$\frac{BF}{EC} = \frac{BC}{CD}$$

$$\frac{BF}{\sqrt{6^2 - 5^2}} = \frac{10}{6}$$

$$BF = \frac{10\sqrt{11}}{6}$$
$$= \frac{5\sqrt{11}}{3}$$

### Marking

## Comments

② argument  
connecting angles  
between the two  
triangles (any method);  
subtract ① per error

① similarity proof  
correct

① correct ratio setup

**1** answer

- connecting arguments and reasoning very poorly communicated, if at all.

- many  
erroneously tried  
to say  $\angle DCE$   
and  $\angle BCF$  were  
equal!

- even if you can't do part (i), assume it is shown and complete part (ii) anyway – many did not do this (this is standard for any HSC-style question).

NOTE: timing – make sure you give some time to every question on the paper – some students clearly “ran out of time”)